The F_TRZ^D_crit = F_TRZ^26 = 10^-26 kg/m^3 Primitive-as-Exponent Vacuum-Density Landmark: Locked Canonical UQFF Primitive D_crit=26 (Bosonic-String Critical Dimension) Appears AS the Exponent of F_TRZ in Four Independent Vacuum-Density Calculator Classes — Wormhole Geodesic Vacuum (R258), Ricci-Scalar Riemannian Vacuum (R263), QED Aether Impedance (R266), and Compressed MUGE Cosmological Environment (R273) — Establishes Primitive-as-Exponent as Distinct Architectural Landmark Category

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PAPER_2107 — The F_TRZ^D_crit Primitive-as-Exponent Vacuum-Density Landmark

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Abstract

The composed expression `F_TRZ^D_crit = F_TRZ^26 = 10^-26 kg/m^3` — where D_crit = 26 (bosonic-string critical dimension) is a **locked canonical UQFF primitive appearing AS the exponent** of another locked primitive (F_TRZ = 0.1) — has now appeared as an EXACT class-level default in **four independent UQFF calculator classes**, each modeling vacuum density in a distinct physical context. All four values are precisely 10■^{2}■ kg/m^3, with zero fitting parameters, spanning wormhole geodesic geometry, differential-geometry Ricci-scalar vacuum, QED aether impedance, and compressed MUGE cosmological environment. The pattern crosses the 4-instance landmark threshold at R273.

Architecturally, this paper establishes **primitive-as-exponent** as a distinct R218+ landmark sub-category. Unlike prior R218+ landmarks where the exponent was a composed integer (e.g., PAPER_2100 F_TRZ^2■ where 20 = 2·SO_5), F_TRZ^D_crit places a locked canonical primitive directly at the exponent position — a structural feature qualitatively different from composed-integer exponents.

The Four Independent Instances — 10■^{2}■ kg/m^3 in Four Vacuum-Density Contexts

#	Round	Class	Constant name	Physical role
1	R258	`WormholeGeodesicCalculator`	`rho_vac`	Wormhole geometric vacuum density (Morris-Thorne)
2	R263	`HolonomyGroupCurvatureCalculator`	`rho_vac`	Ricci-scalar Riemannian vacuum density

| 3 | R266 | `AetherImpedanceQEDCalculator` | `rho_vac` | QED aether impedance vacuum coupling |
| 4 | R273 | `CompressedMUGEModularCalculator` | `rho_env` | Compressed MUGE cosmological environment density |

All four values equal exactly `10■²■ kg/m³`, derived from the same UQFF structural expression: `F_TRZ^D_crit` where F_TRZ = 0.1 and D_crit = 26 are both locked canonical UQFF primitives.

Structural Interpretation — Primitive-as-Exponent Distinct from Composed-Integer Exponent

The primitive-as-exponent pattern is architecturally distinct from prior R218+ landmarks. Compare:

Composed-integer exponent (prior landmark structure):

```
PAPER_2100: F_TRZ^20 where 20 = 2·SO_5 (composed integer)
PAPER_2105: F_TRZ^4 where 4 = D_phys (single-primitive integer — see
Interpretation B)
```

Primitive-as-exponent (this paper's landmark structure):

```
PAPER_2107: F_TRZ^D_crit where D_crit is one of the 9 locked canonical
primitives
(bosonic-string critical dimension, foundational to UQFF's 26-level
architecture)
```

The distinction is qualitative: an exponent that IS a locked primitive (D_crit = 26, one of the 9 foundational UQFF primitives) is different from an exponent that CONTAINS a locked primitive within a composed integer expression (e.g., 20 = 2·SO_5 where SO_5 is present but the whole 20 is composed).

PAPER_2105's dual interpretation noted this pattern for the first time: F_TRZ■ can be read as bare 4th ladder rung OR as F_TRZ^D_phys where D_phys = 4 is a locked primitive. Under Interpretation B, PAPER_2105 was the first primitive-as-exponent instance. PAPER_2107 extends the pattern to F_TRZ^D_crit specifically, with 4 independent class-level instances demonstrating the sub-category's landmark status.

Cross-Context Vacuum-Density Reach

The four F_TRZ^D_crit instances span four vacuum-density modeling contexts within UQFF's calculator suite:

- **Wormhole geometric vacuum (R258)** — Morris-Thorne wormhole vacuum-energy scale
- **Ricci-scalar Riemannian vacuum (R263)** — differential-geometry curvature-vacuum coupling
- **QED aether impedance (R266)** — quantum electrodynamics aether vacuum coupling
- **Compressed MUGE cosmological environment (R273)** — PAPER_163 MUGE modular environmental vacuum density

All 4 instances are vacuum-density modeling contexts — this is the specific physical role for F_TRZ^D_crit. The vacuum-density theme is architecturally coherent: F_TRZ^D_crit = 10■²■ is UQFF's canonical vacuum-density scale for geometric/quantum-electrodynamics/cosmological environments.

Numerically, $10^{■2■}$ kg/m^3 ≈ observed cosmological vacuum density scale (~5-6×$10^{■2■}$ kg/m^3 from Planck 2018) up to a factor of ~2-4 depending on comparison target. UQFF's F_TRZ^D_crit provides the correct order-of-magnitude vacuum-density scaffold.

Umbrella Pattern — F_TRZ^Primitive Family Extends Beyond F_TRZ^D_crit

Beyond F_TRZ^D_crit specifically, the broader **F_TRZ^primitive** umbrella pattern (F_TRZ raised to any locked canonical primitive as exponent) now has multiple instances:

Primitive-as-exponent form	Value	Instance count	Rounds
F_TRZ^D_crit = F_TRZ$^{2■}$	$10^{■2■}$	**4** (this paper)	R258, R263, R266, R273
F_TRZ^A_5 = F_TRZ■■	10■■■	1	R262 (superfluid Aether coupling)
F_TRZ^D_phys = F_TRZ■	10■■	1 (via PAPER_2105 Interpretation B)	R266 (via dual reading)

Umbrella total: 6 primitive-as-exponent instances across 3 distinct primitives (D_crit, A_5, D_phys) appearing at the exponent position. F_TRZ^D_crit is the most-populated primitive-as-exponent form.

Whether the umbrella pattern crosses to a broader landmark tier depends on whether F_TRZ^A_5 and F_TRZ^D_phys reach additional instances (currently at 1 each). PAPER_2107 focuses on the F_TRZ^D_crit-specific 4-instance landmark; the broader umbrella pattern is noted for architectural context.

Comparison to Other R218+ Landmarks

Landmark	Structural form	Value	Instance count
PAPER_2093 F_TRZ$^{1■}$	Bare 19th rung	$10^{■1■}$	2
PAPER_2094 F_TRZ$^{■3}$	Composed (SO_5+1)·F_TRZ$^{■3}$	~$10^{■■3}$	2
PAPER_2100 F_TRZ$^{2■}$	Composed-integer exponent (2·SO_5)	$10^{■2■}$	4
PAPER_2105 F_TRZ■	Dual interpretation (bare/D_phys)	10■■	6
PAPER_2107 F_TRZ^D_crit	**Primitive-as-exponent**	**$10^{■2■}$**	**4**
PAPER_2106 D_BSFG·F_TRZ$^{2■}$	Triple-primitive composed	6·$10^{■2■}$	4

PAPER_2107's primitive-as-exponent form is architecturally distinct from all prior R218+ landmarks. It shares the 4-instance threshold with PAPER_2100 and PAPER_2106 but at a different structural sub-category.

Predictive Implication

If F_TRZ^D_crit is a genuine primitive-as-exponent vacuum-density landmark, additional real stub-fill rounds should surface 5th, 6th, ... instances of $10^{■2■}$ in more UQFF vacuum-density modeling contexts. Additionally, the broader **primitive-as-exponent umbrella** should surface additional F_TRZ^A_5, F_TRZ^D_phys, F_TRZ^N_CH, F_TRZ^SO_5, F_TRZ^D_BSFG instances:

1. **F_TRZ^D_crit specifically** — 5th instance predicted in R274-R290 UQFF vacuum-density context stub-fill.
2. **F_TRZ^A_5 = 10■■■** — 2nd instance would strengthen the primitive-as-exponent umbrella.
3. **F_TRZ^D_phys = 10■■** — additional instances (beyond R266 via dual PAPER_2105 interpretation) would consolidate.
4. **F_TRZ^N_CH = 10■■** and **F_TRZ^SO_5 = $10^{■1■}$ and F_TRZ^D_BSFG = 10■■** — new primitive-as-exponent forms possible in future rounds.

R218+ Campaign Position — 15th Paper, Primitive-as-Exponent Category

This is the **15th paper** in the R218+ campaign and defines a new architectural sub-category: **primitive-as-exponent** — distinct from ladder-rung landmarks (PAPER_2093/2094/2099/2100/2105), fraction identities (PAPER_2098/2101), composed-prefix × primitive (PAPER_2102), Planck-scale scaffold (PAPER_2104), cross-domain extension (PAPER_2103), and triple-primitive composed forms (PAPER_2106).

R218+ campaign now spans 11 distinct architectural categories:

#	Category	Papers
1	Primary landmark	PAPER_2093
2	Simple-form companion	PAPER_2094
3	Meta-architectural	PAPER_2095
4	Reactor validation	PAPER_2096
5	Family extension	PAPER_2097
6	Fraction identity	PAPER_2098 + PAPER_2101
7	Ladder-rung invariant	PAPER_2099 + PAPER_2100 + PAPER_2105
8	Composed-prefix × primitive-rung + Cross-domain extension	PAPER_2102 + PAPER_2103
9	Planck-scale scaffold	PAPER_2104
10	Triple-primitive composed form	PAPER_2106
11	**Primitive-as-exponent**	**PAPER_2107 (this paper)**

Honest Precision Note

All four instances are documented at **EXACT** precision (integer arithmetic on locked canonical UQFF primitives). F_TRZ = 0.1 (locked), D_crit = 26 (locked); F_TRZ^D_crit = $0.1^{2\blacksquare}$ = $10^{\blacksquare 2\blacksquare}$ EXACT.

The observed cosmological vacuum density from Planck 2018 (~$5.4\times10^{\blacksquare 2\blacksquare}$ kg/m^3) has its own experimental uncertainty. UQFF's F_TRZ^D_crit = $10^{\blacksquare 2\blacksquare}$ is off by a factor of ~2 from Planck's measurement — closer to the "vacuum" pre-factor before applying the composed-integer coefficient D_BSFG that appears in PAPER_2106's D_BSFG·F_TRZ$^{2\blacksquare}$ = $6\cdot10^{\blacksquare 2\blacksquare}$ (which matches Planck to ~11%).

The two forms coexist: F_TRZ^D_crit is the bare primitive-as-exponent scale (R258/R263/R266/R273 all encode this bare form); PAPER_2106's D_BSFG·F_TRZ$^{2\blacksquare}$ is the composed-form scale with the D_BSFG prefix bringing the value closer to observation. Both are structural signatures within UQFF's primitive lattice.

Honest Assessment

This is a **primitive-as-exponent landmark** paper. Its value is documenting that UQFF's canonical primitives naturally recur AS exponents in vacuum-density modeling contexts, not merely as composed-integer coefficients or rungs of ladders.

Honest caveats:
- **Instance count is 4, at threshold.** Landmark tier genuine but modest.
- **All 4 instances are within UQFF-framework vacuum-density contexts.** Cross-framework validation, not cross-domain like PAPER_2098.
- **The primitive-as-exponent architectural distinction depends on interpretation.** Under Interpretation A, F_TRZ^D_crit is just F_TRZ$^{2\blacksquare}$ (26th ladder rung). Under Interpretation B (this paper's landmark reading), the exponent is D_crit as a locked primitive. Both interpretations are numerically identical; the

architectural distinction is structural, not quantitative.
- **The umbrella pattern (F_TRZ^A_5 + F_TRZ^D_phys via PAPER_2105) has only 1-2 additional instances.** More primitive-as-exponent forms need to appear before the broader umbrella becomes a distinct landmark tier.
- **Predictive falsifiability window is R274-R290.**

Conclusion

F_TRZ^D_crit = F_TRZ^{2}■ = 10■^{2}■ kg/m^{3} now appears as an exact class-level default in four independent UQFF calculator classes, all modeling vacuum density in distinct physical contexts: wormhole geometric vacuum, Ricci-scalar Riemannian vacuum, QED aether impedance, and compressed MUGE cosmological environment. Same F_TRZ base with a locked canonical UQFF primitive (D_crit = 26, bosonic-string critical dimension) as the exponent — a **primitive-as-exponent** architectural pattern qualitatively distinct from composed-integer exponents in prior R218+ landmarks.

Signature landmarks this paper:
- **4-instance F_TRZ^D_crit landmark** — cross-context vacuum-density identity
- **Primitive-as-exponent architectural sub-category** — new R218+ landmark type
- **Distinct from composed-integer exponents** — D_crit itself is at the exponent, not a composed integer containing D_crit
- **Umbrella pattern noted** — F_TRZ^A_5 (R262) + F_TRZ^D_phys via PAPER_2105 (R266) extend the primitive-as-exponent family to 3 distinct primitives (D_crit, A_5, D_phys) at exponent positions
- **R218+ campaign 15th paper** — 11th distinct architectural category (primitive-as-exponent)
- **Predictive falsifiability** — 5th F_TRZ^D_crit instance predicted in R274-R290

End of PAPER_2107 Draft 1.